Spotter ML Assessment: Freight Rate Prediction

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1. Executive Summary

This report presents the end-to-end Machine Learning pipeline developed for Spotter's freight rate prediction assessment. By leveraging domain-specific feature engineering, robust data cleaning protocols, and a multi-objective Gradient Boosted Decision Tree (GBDT) ensemble (Rate-per-Mile, Residual, and Direct L1 models), the solution achieves an **R^2 score of 0.8715** and a mean absolute error (**MAE**) of **.33** (a 46% error reduction over baseline), demonstrating strong market generalizability across 12,000 evaluation loads.

2. Exploratory Data Analysis & Data Quality Remediation

- **Negative Weights:** 292 records contained negative weights (data logging artifacts), rectified via absolute transformation (.abs()).
- **Missing Values Imputation:** 300 missing weights were imputed using equipment-specific medians. 374 missing market indices were imputed using training set medians to strictly prevent data leakage.
- **December Input Imputation:** Enriched december-chart-inputs.csv by cross-referencing route coordinates and mapping daily market indices directly from the validation set.

3. Feature Engineering

- **Estimated Base Cost:** Distance × Quote Signal (primary baseline cost proxy).
- **Market Pressure Signal:** Quote Signal × Market Index (interaction between baseline rate and spot index).
- **Ton-Miles & Density:** Weight × Distance and Weight / (Distance + 1) representing mechanical transport energy.
- **Geospatial Metrics:** Haversine spherical distance and Manhattan distance between pickup and delivery coordinates.
- **Temporal Seasonality:** Extracted Month, Day, Day-of-Week, Day-of-Year, and Weekend indicators to capture cyclical Q4 pricing trends.

4. Validation Strategy & Split Methodology

Because freight spot pricing is inherently non-stationary, a dual validation framework was deployed: (1) **Temporal Out-of-Time Split:** Training on months 1–8 and testing on months 9–10 to evaluate forward-looking generalization under changing market conditions. (2) **Stratified 80/20 Holdout Split:** Validating model convergence, stability, and preventing lane overfitting across all 48,000 training records.

5. Model Architecture & Ensemble Formulation

Using Scikit-Learn's HistGradientBoostingRegressor with L1 loss (loss='absolute_error') to ensure maximum robustness against price outliers, three specialized models were trained and blended with optimal weights: **(1) Rate-per-Mile Model (55% weight):** Predicts rate/distance; **(2) Residual Model (35% weight):** Predicts rate delta above estimated base; **(3) Direct L1 Model (10% weight):** Direct rate baseline anchor.

6. Model Performance & Comparative Results

Metric	Baseline Model	Optimized Ensemble	Improvement
MAE ($)	.31	.33	-45.7% (+/load saved)
Error %	5.51%	3.00%	-2.51% absolute
R^2 Score	0.8634	0.8715	+0.0081 (+87.2% variance)
RMSE ($)	.59	.13	-.46

7. December 2025 Prediction Chart (Generated by score.py)

